EE3001: Solid State Devices
TAs: Alias George (ee24d402@smail.iitm.ac.in) and Devendar K (ee24s015@smail.iitm.ac.in)
Assignment #6

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10}cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$3.2 \times 10^{19}cm^{-3}$, N_v=$1.8 \times 10^{19}cm^{-3}$, μ_n = $1350cm^2/V.s$, μ_p = $450cm^2/V.s$; Permittivity for vacuum: ϵ_0=8.85×10^{-14}F/cm, relative permittivity: $\epsilon_{r,Si}$=11.8, ϵ_{r,SiO_2}=3.9; pn diode dopants are completely ionized.

1. (2 points) The conductance (g) versus forward biased current (I) plot for a pn diode with unity ideality factor yields the y-intercept and slope as 38.46×10^{-12} S and 38.46 V^{-1}, respectively.

   (a) Find out the bias voltage at which this diode yields 2 mA current.

   (b) What is the diode resistance at this bias point?

2. (2 points) Given two identically and uniformly doped abrupt silicon p-n junction diodes, D1 and D2. Cross sectional area of D1 is twice of that of D2. The reverse saturation current for D1 is 10 nA.

   (a) Plot the forward current dependent conductance for both the diodes at room temperature (in the same figure) showing important projection points.

   (b) If both D1 and D2 were p^+-n diodes, plot their forward current dependent diffusion capacitance (in the same figure) if τ_p= 1 μs for both the diodes.

3. (5 points) Qualitatively draw the position (x) dependent electron and hole current densities ($J_n(x)$ and $J_p(x)$) for the following p-n diode structures where W_p, W and W_n represent the lengths of the quasi-neutral p-region, space-charge-region, and quasi-neutral n-region, respectively. Note that L_p and L_n denote the diffusion lengths for minority holes and electrons, respectively.

   (a) A p-n diode with $N_A > N_D$, $W_n \gg L_p$ and $W_p \gg L_n$.

   (b) A p-n diode with $N_A > N_D$, $W_n \ll L_p$ and $W_p \ll L_n$.

   (c) A p-n diode with $N_A > N_D$, $W_n \gg L_p$ and $W_p \ll L_n$.

   (d) A p^+-n diode with a $W_n \gg L_p$.

   (e) A p^+-n diode with a $W_n \ll L_p$.

1. (2 points) The conductance (g) versus forward biased current (I) plot for a pn diode with unity ideality factor yields the y-intercept and slope as 38.46×10^{-12} S and 38.46 V^{-1}, respectively.

   (a) Find out the bias voltage at which this diode yields 2 mA current.

   (b) What is the diode resistance at this bias point?

$I = I_0 (e^{qV/KT} - 1)$ $\quad g = \frac{dI}{dV} = \frac{qI_0}{KT} e^{qV/KT}$

$\Rightarrow g = \left(\frac{q}{KT}\right) I + \left(\frac{qI_0}{KT}\right)$

$\frac{KT}{q} = 0.026\,V \Rightarrow T = 300K$

$I_0 = 10^{-12} A$

$2 \times 10^{-3} = 10^{-12} (e^{V/V_T} - 1) \approx 10^{-12} e^{V/V_T} \Rightarrow V = 0.557$

$g = 38.46 (2 \times 10^{-3}) + (38.46 \times 10^{-12})$

$\Rightarrow R = 13\,\Omega$

2. (2 points) Given two identically and uniformly doped abrupt silicon p-n junction diodes, D1 and D2. Cross sectional area of D1 is twice of that of D2. The reverse saturation current for D1 is 10 nA.

   (a) Plot the forward current dependent conductance for both the diodes at room temperature (in the same figure) showing important projection points.

   (b) If both D1 and D2 were p^+-n diodes, plot their forward current dependent diffusion capacitance (in the same figure) if τ_p= 1 μs for both the diodes.

$I = I_S (e^{V/V_T} - 1)$

$= \frac{Q_p}{\tau_p} + \frac{Q_n}{\tau_n}$

$(I_0 \propto A)$

$g = \frac{dI}{dV} = \frac{q(I+I_0)}{KT} = \frac{1}{V_T}(I + I_0)$ $\quad$ Slope $= \frac{1}{V_T} = 38.46\,V^{-1}$

a) $\frac{I_{01}}{V_T} = \frac{10nA}{26mV} = 384.6\,nS$ $\quad \frac{I_{02}}{V_T} = \frac{1}{2}\left(\frac{I_{01}}{V_T}\right) = 192.3\,nS$

b) p^+n diode $\Rightarrow$ C_{diff} is dominated by holes storage in n-region $\quad I \approx \frac{Q_p}{\tau_p} \Rightarrow Q_p = \tau_p I \Rightarrow \frac{dQ_p}{dV} = C_d = \tau_p \frac{dI}{dV}$

$C_d = \frac{\tau_p I_0}{V_T} + \left(\frac{\tau_p}{V_T}\right) I$

4. (5 points) For a forward biased p^+-n diode, the hole current density ($J_p(x)$) at $x = x_n$ is found to be 0.1 mA/cm^2. Given that $\tau_p = 1$ μs.

If the length of the quasi-neutral n-region is 300 μm, answer the following:

(a) What is the total diode current density at the given bias?

(b) What is the electron current density, $J_n(x)$ at $x = x_n + 2$ μm?

(c) What is the total stored excess hole charge in the quasi-neutral n-region?

If the length of the quasi-neutral n-region is 2 μm, answer the following:

(d) What is the total stored excess hole charge in the quasi-neutral n-region?

(e) What is the electron current density, $J_n(x)$ at $x = x_n + 1$ μm?

$N_A >> N_D$

W_n

$\tau_T = \frac{(2\times10^{-4})^2}{2\times 11.7} = 1.71\text{ns}$

$L_p = \sqrt{D_p \tau_p} \approx 31.6\mu m$

$W_n >> L_p$ (long base diode).

a) $J_{total} \approx J_p(x_n) = 0.1\ mA/cm^2$

b) $J_p(x_n + 2\mu m) = 0.1\times10^{-3} \times e^{-2/31.6} = 0.0939\ mA/cm^2$

$\Rightarrow J_n(2\mu m) = 0.1 - 0.0939 = 0.061\ mA/cm^2$

c) $Q_p = J_p(x_n)\,\tau_p = 10^{-10}\ C/cm^2$

$W_n << L_p$ (short-base diode)

d) $\tau_T = \frac{W_n^2}{2D_p} = 1.71ns \Rightarrow Q_p = J_p(x_n)\,\tau_p = 1.71\times10^{-13}\ /cm^2$

4. (5 points) For a forward biased p^+-n diode, the hole current density ($J_p(x)$) at $x = x_n$ is found to be 0.1 mA/cm^2. Given that τ_p = 1 μs.

   If the length of the quasi-neutral n-region is 300 μm, answer the following:

   (a) What is the total diode current density at the given bias?

   (b) What is the electron current density, $J_n(x)$ at $x = x_n + 2$ μm?

   (c) What is the total stored excess hole charge in the quasi-neutral n-region?

   If the length of the quasi-neutral n-region is 2 μm, answer the following:

   (d) What is the total stored excess hole charge in the quasi-neutral n-region?

   (e) What is the electron current density, $J_n(x)$ at $x = x_n + 1$ μm?

3. (5 points) Qualitatively draw the position (x) dependent electron and hole current densities ($J_n(x)$ and $J_p(x)$) for the following p-n diode structures where W_p, W and W_n represent the lengths of the quasi-neutral p-region, space-charge-region, and quasi-neutral n-region, respectively. Note that L_p and L_n denote the diffusion lengths for minority holes and electrons, respectively.

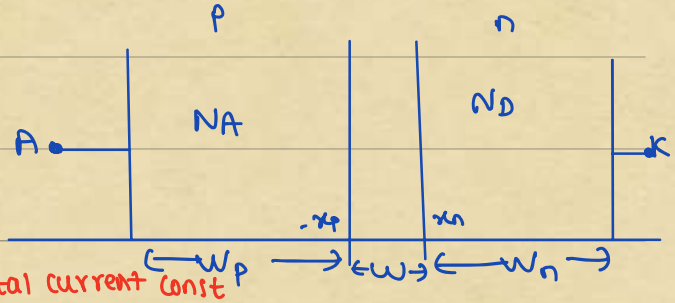
(a) A p-n diode with $N_A > N_D$, $W_n \gg L_p$ and $W_p \gg L_n$. → both sides long

(b) A p-n diode with $N_A > N_D$, $W_n \ll L_p$ and $W_p \ll L_n$. → both sides short

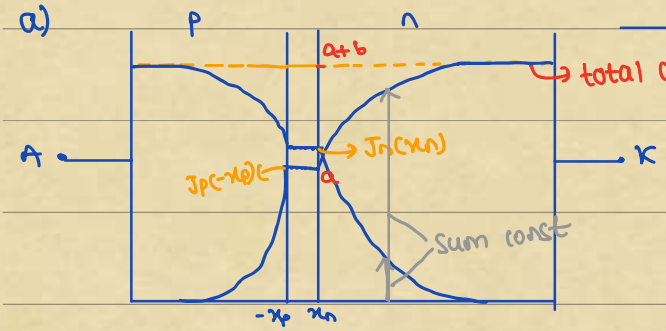
(c) A p-n diode with $N_A > N_D$, $W_n \gg L_p$ and $W_p \ll L_n$. → n long p short

(d) A p^+-n diode with a $W_n \gg L_p$. → long

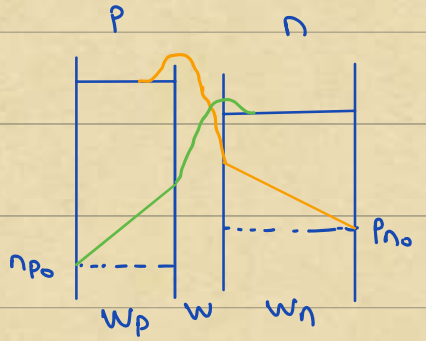
(e) A p^+-n diode with a $W_n \ll L_p$. → short



a)

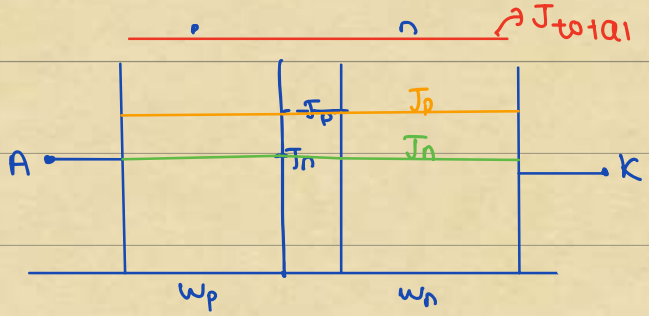


b)

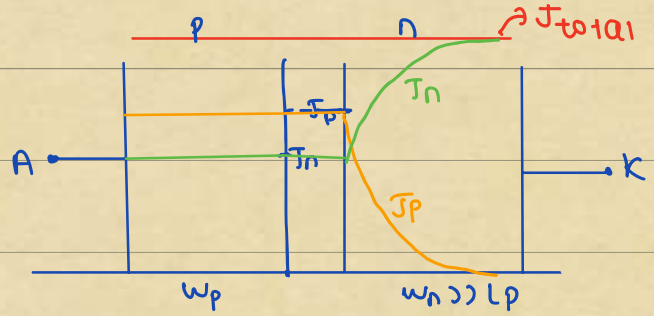


$$\frac{d^2 \Delta p_n(x)}{dx^2} = \frac{\Delta p_n(x)}{\tau_p^2} = 0$$ (no generation or recombination)

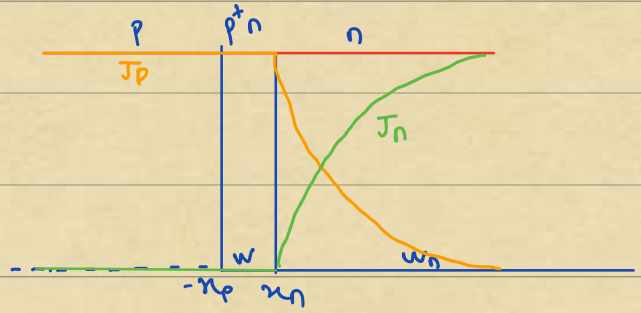
from continuity



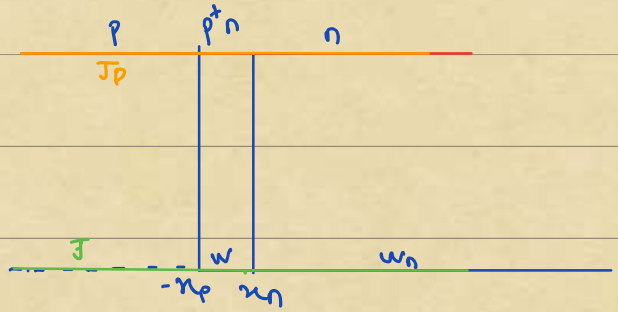
c)


p
n
J_{total}
J_n
J_p
A
K
J_p
J_n
w_p
w_n

d)


p
p^+n
n
J_p
J_n
w
w_n
$-x_p$
x_n

e)


p
p^+n
n
J_p
J
w
w_n
$-x_p$
x_n

EE3001: Solid State Devices
TAs: Alias George (ee24d402@smail.iitm.ac.in) and Devendar K (ee24s015@smail.iitm.ac.in)
Assignment #7

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10} cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$3.2 \times 10^{19} cm^{-3}$, N_v=$1.8 \times 10^{19} cm^{-3}$, $\mu_n = 1350 cm^2/V.s$, $\mu_p = 450 cm^2/V.s$; Permittivity for vacuum: ϵ_0=8.85×10^{-14}F/cm, relative permittivity: $\epsilon_{r,Si}$=11.8, ϵ_{r,SiO_2}=3.9; pn diode dopants are completely ionized.

1. (2 points) At T=300 K, a p^+n junction with a long n region and reverse saturation current of $I_0 = 1\ \mu A$, is carrying a steady forward current $I_1 = 1$ mA for $t < 0$. At $t = 0$, the diode current is abruptly increased to $I_2 = 10$ mA. Calculate the diode voltage at $t = 0.5\ \mu s$ if $\tau_p = 1\ \mu s$.

2. (5 points) In switching applications, a diode has to be switched from forward to reverse bias condition and back. Assume that the diode under investigation is a p^+n junction diode with a long n region with a reverse saturation current of $I_0 = 1\ \mu A$ and is operating at 300 K. The switching diode is connected in series with a load resistor R=1 kΩ. The diode and the load resistor is driven by a voltage source that switches from +10 V to -5 V with a low enough switching frequency. The state of the voltage source is +10 V allowing a steady forward current $I_1 = 10$ mA through the diode for $t < 0$. At $t = 0$, the diode is abruptly reverse biased to -5 V. Now the stored minority carrier will gradually deplete; hence, will take some time to rearrange themselves according to the external bias condition. Given that $\tau_p = 1\ \mu s$.

   (a) At what time (known as storage delay time) will the stored minority charge become zero?

   (b) What is the final diode voltage and current?

   (c) If the voltage source switches between +5 V to -10 V, what would be the value of this time constant calculated in (a)?

   (d) If the diode under consideration corresponds to a p^+-n-n^+ structure with the quasi-neutral n-region length as 1 μm, find out t_{sd} for this device if the voltage source switches between +10 V to -5 V and between +5 V to -10 V.

3. (5 points) In a p^+-n-n^+ diode, the doping concentration of the n-region is 2×10^{15} cm^{-3}. If the critical field at avalanche breakdown is 0.2 MV/cm, answer the folowing.

1. (2 points) At T=300 K, a p^+n junction with a long n region and reverse saturation current of $I_0 = 1\ \mu A$, is carrying a steady forward current $I_1 = 1$ mA for $t < 0$. At $t = 0$, the diode current is abruptly increased to $I_2 = 10$ mA. Calculate the diode voltage at $t = 0.5\ \mu s$ if $\tau_p = 1\ \mu s$.

$$V = V_T \ln\left(\frac{I_1 e^{-t/\tau_p} + I_2(1 - e^{-t/\tau_p})}{I_0}\right)$$

3. (5 points) In a p^+-n-n^+ diode, the doping concentration of the n-region is 2×10^{15} cm^{-3}. If the critical field at avalanche breakdown is 0.2 MV/cm, answer the folowing.

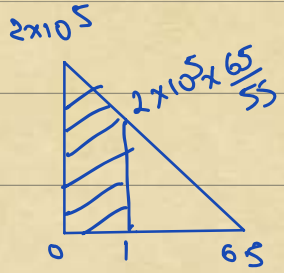
(a) What will be the breakdown voltage of the diode if the width of the n-region is 10 μm? Sketch the electric field distribution.

(b) What will be the breakdown voltage of the diode if the width of the n-region is 1 μm? Sketch the electric field distribution.

$$2\times10^5 = \frac{1.6\times10^{-19}\times 2\times10^{15} W}{10^{-12}} \Rightarrow W = 6.5\mu m$$

a) $V_{BR} = \frac{1}{2}\times 2\times10^5 \times 6.5\mu m = 65V$

b) $\frac{1}{2}\times 2\times10^5\left(\frac{120}{65}\right) = 18.461\ V$



(a) What will be the breakdown voltage of the diode if the width of the n-region is 10 μm? Sketch the electric field distribution.

(b) What will be the breakdown voltage of the diode if the width of the n-region is 1 μm? Sketch the electric field distribution.

EE3001: Solid State Devices
TAs: Alias George (ee24d402@smail.iitm.ac.in) and Devendar K (ee24s015@smail.iitm.ac.in)
Assignment #8

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10} cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$3.2 \times 10^{19} cm^{-3}$, N_v=$1.8 \times 10^{19} cm^{-3}$, μ_n = $1350 cm^2/V.s$, μ_p = $450 cm^2/V.s$; Permittivity for vacuum: ϵ_0=8.85×10^{-14}F/cm, relative permittivity: $\epsilon_{r,Si}$=11.8, ϵ_{r,SiO_2}=3.9; pn diode dopants are completely ionized.

1. (2 points) At a particular wavelength of incident radiation, the absorption coefficients of Si and GaAs are 4000 per cm and 30,000 per cm, respectively. What are the thickness of Si and GaAs necessary to absorb 80% of the incident radiation?
   $0.2 = e^{-\alpha x} \Rightarrow \alpha x = 1.609 \Rightarrow x = 4\mu m$ for Si & $0.536\mu m$ for GaAs
2. (5 points) A 5 μm thick Si sample is illuminated with monochromatic light having energy of 2 eV. The incident power is 10 mW.
   $P_{absorbed} = 10^{-2}(1 - e^{-4000 \cdot 5 \times 10^{-4}})$
   (a) What is the total energy absorbed in the sample per second? $= 8.647$ mW
   (b) How much energy per second is dissipated as heat? frac as heat $= \frac{h\nu - E_g}{h\nu} = 1 - \frac{1.12}{2} = 0.44$
   (c) How many EHPs are generated in the sample per second? $G = P_{abs}/h\nu = 8.647 \times 10^{-3}/2 \times 1.6 \times 10^{-19} = 2.7 \times 10^{16}$ EHPs $\Rightarrow 3.805$ mV as heat
3. (5 points) A solar cell has an area of 1 cm $\times$ 1 cm. The cell has an n^+p structure, where the doping concentration of the p-region is 10^{15} $/cm^3$. Considering μ_n = 1350 cm^2/V-sec and τ_n = 1 μs in the p-region, and I_L = 40 mA, calculate I_{sc}, V_{oc}, maximum power output (P_m) and the Fill Factor (FF). Estimate the incident optical power if the efficiency of the solar cell is 15 %.

4. (15 points) This problem will help us understand how we can derive maximum power from a solar cell. Also, it is important to adjust the load in order to obtain maximum power at different conditions, viz. sunny or cloudy; morning, noon or evening.

   Background information: To generate power from a p-n solar cell, a load resistor (R) can be connected across it. The current (I) flowing through the solar cell also flows through R. The voltage across the resistor V=-IR also appears across the diode, which is forward biased by the same voltage (V). Hence, the current through the solar cell consists of two components: (i) forward biased current from p to n, and (ii) light generated current (I_L) from n to p. Total current is given as

3. (5 points) A solar cell has an area of 1 cm × 1 cm. The cell has an n^+p structure, where the doping concentration of the p-region is 10^{15} /cm^3. Considering μ_n = 1350 cm^2/V-sec and τ_n = 1 μs in the p-region, and I_L = 40 mA, calculate I_{sc}, V_{oc}, maximum power output (P_m) and the Fill Factor (FF). Estimate the incident optical power if the efficiency of the solar cell is 15 %.

$N_A = 10^{15}$ $A = 1\,cm^2$ $\mu_n = 1350\,cm^2/Vs$ $\tau_n = 10^{-6}s$ $I_L = 40mA$

$I_{SC} = I_L = 40mA = 4\times10^{-2}A$

$$I_S = \frac{qAD_n n_i^2}{L_n N_A} = \frac{qA\mu_n V_T n_i^2}{\sqrt{D_n \tau_n}\, N_A} = 9.5\times10^{-11}\,A$$

$$V_{OC} = V_T \ln\left(\frac{I_L}{I_S}+1\right) = 0.516\,V$$

$V_{OC} = 19.85\,V_T$

$$FF = \frac{19.85 - \ln(19.85+0.72)}{19.85+1} \approx 0.807$$

$P_m = F.F.\,V_{OC}.\,I_{SC} = 16.6mV$

$\Rightarrow P_{in} = 0.111W$

$$I = I_0(exp(V/V_T) - 1) - I_L \quad (1)$$

and the power generated by the solar cell (and dissipated across R) can be given as

$$P = -IV \quad (2)$$

Problem: The reverse saturation current (I_0) in a p-n junction solar cell is 100 pA.

(a) The solar cell is taken out in the sun. The incident light results in a light generated short-circuit current ($I_{sc} = I_L$) of 40 mA through the solar cell. For this condition of sunlight,

(i) what would be the open circuit voltage (V_{oc}) across the two terminals of the p-n diode?

(ii) what would be the maximum possible power (P_m) which can be delivered by the solar cell to a load resistor connected across it?

(iii) what is the fill factor (FF=$\frac{P_m}{V_{oc}I_{sc}}$) of the solar cell?

(iv) what is the value of R (let's call it $R_{m40} = -\frac{V_m}{I_m}$) required to obtain maximum power across the load?

(b) After some time it became cloudy and I_L is reduced to 10 mA. For this condition of sunlight,

(i) what is the open circuit voltage (V_{oc})?

(ii) what would be the maximum possible power (P_m) which can be delivered by the solar cell to a load resistor connected across it?

(iii) what is the fill factor (FF) of the solar cell?

(iv) what is the value of R (let's call it R_{m10}) required to obtain maximum power across the load?

(c) Suppose we did not change the load resistance and we had $R = R_{m40}$ when $I_L = 10mA$.

(i) Draw the I-V characteristics of the solar cell for the two conditions, i.e., for $I_L = 40$ mA and $I_L = 10$ mA. Draw a straight line (called the load line) corresponding to $V = -IR_{m40}$. The operating point has to be the point of intersection of the load line and the I-V characteristics. Show that with $I_L = 10$ mA and $R = R_{m40}$, $I = -I_L$. This can be verified by calculating V if $I = -I_L$ and substituting this value of V in eq. (1).

(ii) What is the power delivered by the solar cell?

(iii) What is the fraction of power that is available from the solar cell but is wasted by not changing the load resistance from R_{m40} to R_{m10}?

EE3001: Solid State Devices
TAs: Alias George (ee24d402@smail.iitm.ac.in) and Devendar K (ee24s015@smail.iitm.ac.in)
Assignment #9

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following value: k_BT=0.026 eV (at room temperature)

1. (6 points) A metal-semiconductor Schottky diode (MS1) fabricated on n-GaAs with $N_D = 10^{16}\ cm^{-3}$ has a reverse saturation current density $J_s = 5 \times 10^{-8}$ A/cm^2 at T=300 K. Given $N_c = 4.7 \times 10^{17}\ /cm^3$ and $\epsilon_r = 13.1$ for GaAs.

   (a) Calculate the barrier height (both from metal-to-semiconductor) and semiconductor-to-metal) and the depletion region width at thermal equilibrium assuming the effective Richardson's constant $A^{**} = C_1 N_c / T^2 = 4.4\ A/K^2cm^2$.

   (b) Calculate the minority carrier diffusion current density (assuming this as a p^+n junction) and show that the Schottky diode current density is mostly due to majority carriers (electrons). Assume the hole diffusion coefficient to be 10 cm^2/sec and minority carrier lifetime as 10^{-8} sec.

2. (2 points) We want to design a BJT with β = 150. Consider that for this BJT, base transport factor (α_T) is 0.995. If $\mu_n/\mu_p = 3$ and $W_B/W_E = 0.5$, and all regions are short, what is the required value of N_{DE}/N_{AB}?

3. (2 points) Two transistors T1 and T2 are identical in all respects except in their base widths. The emitter efficiencies are unity in both cases. The base width of T1 is 1 μm and that of T2 is 2 μm. The transistors are biased in the active region of operation. The base current of T1 is 10 μA when the collector current is 1 mA. Determine the base current of T2 when the collector current is 1 mA.

4. (2 points) The β of an n^+pn^+ transistor is 100 when the collector-base junction is reverse biased with $V_{CB} + V_{bi}$ = 1 V. The neutral base width, that is, the undepleted portion of the base for this case is 1.2 μm and the depletion layer width at the CB junction is 0.3 μm. Determine the β of the transistor when $V_{CB} + V_{bi}$ = 9 V considering the following two cases separately: (a) $\alpha_T = 1$ and $\gamma \neq 1$ (b) $\alpha_T \neq 1$ and $\gamma = 1$. Note that V_{CB} is the reverse voltage applied at the CB junction while V_{bi} is the built-in potential of this junction.